Problem B

Predicting Penalty Trajectories: What Two Early Positions Can Resolve

Abstract

We predict where a penalty kick will cross the goal plane using positions at 50 and 100 ms. Under centimetre rounding, exact release timing and a 3 m/s horizontal wind cap, the prescribed hypothetical kick has its most probable goal cell centred at approximately $(2.56, 2.07)$ m. Six cells cover 0.3572 m^2 and contain 90.52% of the model probability conditional on direct entry. The median crossing time is 0.399 s. Two admissible flights match the same early readings and cross 8.76 cm apart, giving a conditional 4.38 cm worst-case lower bound on point error.

To test the general method at 100 ms, we calibrate regions on 90 synthetic kicks and evaluate 120 separate kicks, including 71 direct entries. With centimetre rounding alone, both aerodynamic forecasts cover 70/71 entries at their 95% calibration settings. The two-reading region averages 0.585 square metres at that observed coverage, compared with 3.482 for the late reading and 1.222 for gravity. The two readings also reduce mean point error by 3.80 cm relative to the late reading. With 6 mm independent and 8 mm shared tracking errors, gravity uses 0.933 square metres against 1.385 for the two-reading forecast at an observed coverage of 67/71. The matched operating points show that the advantage depends on measurement quality. The model outputs explicit event probabilities and flags unsupported predictions. Real-match cases provide timing diagnostics. Release-synchronized field accuracy remains open.

1 Flight geometry in the first 100 milliseconds

By the second observation, the ball in the prescribed example has travelled 2.96 m toward a goal plane 11 m from the penalty spot. The two readings constrain its mean velocity over 50 ms. The remaining 8.04 m make the goal crossing sensitive to forces that the early positions constrain only partially. The contest asks for the most likely point of entry, the uncertainty in both coordinates and a test of the forecast [1].

The output is a point and a joint region on the goal plane, with a separate probability of direct entry. The prescribed example is hypothetical and provides the input positions only. Figure 1 links the observations to the prediction, and Table 1 locates the evidence for each requested result.

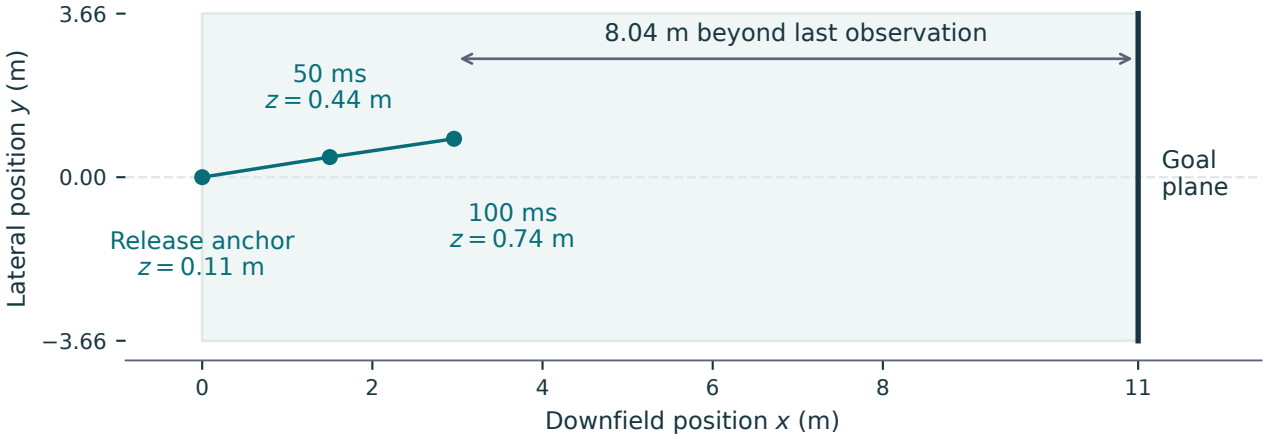


Figure 1. The data end 8.04 m before the goal plane. Plan view of the launch anchor and the two prescribed positions, with height printed beside each marker. The straight segments join the inputs as guides. Downfield and lateral axes have different visual scales.

Table 1. Problem requirements and corresponding evidence. The predictions in the first and second rows use only information available at 100 ms.

Task	Quantity returned or checked	Evidence
Develop a predictor	First-event probabilities and highest-mass goal cell	Eqs. (3), (5)
Estimate uncertainty	Joint region and sensitivity to observation and force assumptions	Figs. 3, 4
Apply and test	Prescribed case plus a common-deadline synthetic comparison	Eq. (6), Fig. 6

1.1 Release geometry and first-event outcomes

Take x toward the goal, y across its face from the centre, and z upward from the ground. The ball centre starts at $\boldsymbol{r}_0 = (0, 0, 0.11)$ m. The prescribed positions, in metres, are

$$\boldsymbol{r}(0.050\,\mathrm{s}) = (1.50, 0.45, 0.44), \qquad \boldsymbol{r}(0.100\,\mathrm{s}) = (2.96, 0.86, 0.74). \tag{1}$$

Our nominal analysis takes the release position and times as exact. The problem does not specify a measurement-error distribution, a ball type, wind or spin. We therefore interpret the displayed decimal precision as centimetre rounding for a reproducible reference scenario, and test alternatives explicitly. Each later coordinate q, displayed to the nearest centimetre, admits the half-open interval

$[q-\epsilon, q+\epsilon)$, where $\epsilon = 0.005$ m. Tracking error and release-time uncertainty enter the observation layer developed below. These are alternative conditions, not measured properties of the hypothetical kick.

We follow the ball until its centre crosses $x = 11$ m or descends to one ball radius above ground. A crossing through the opening, $-3.66 < Y < 3.66$ m and $0 < Z < 2.44$ m, is a direct entry G. This ball-centre criterion neglects collisions with the frame, rebounds and goalkeeper interventions. The other outcomes are a crossing outside the opening, earlier ground contact, or neither event by 1.25 s. We estimate all four probabilities and the joint distribution of $Q = (Y, Z)$ conditional on G.

1.2 A lower bound on wind speed under drag alone

The mean horizontal velocities in the two successive intervals are $(30.0, 9.0)$ and $(29.2, 8.2)$ m/s. With drag alone and constant horizontal wind $\boldsymbol{w}$, the horizontal heading in the air frame remains fixed. The displacements $\boldsymbol{p}_i$ from release must therefore satisfy

$$\det(\boldsymbol{p}_1 - t_1\boldsymbol{w},\, \boldsymbol{p}_2 - t_2\boldsymbol{w}) = 0, \qquad \|\boldsymbol{w}\| \geq \frac{|\det(\boldsymbol{p}_1, \boldsymbol{p}_2)|}{\|t_2\boldsymbol{p}_1 - t_1\boldsymbol{p}_2\|}. \tag{2}$$

The determinant is the signed area spanned by two planar vectors. The displayed coordinates require at least 14.849 m/s of wind. At fixed wind, the determinant is linear in each coordinate separately and attains its extrema over the closed rounding boxes at their corners. Across the half-open rounding boxes, 4.353 m/s is the infimum wind speed compatible with horizontal collinearity. Our 3 m/s wind cap therefore rules out a drag-only explanation. This is a necessary geometric condition, not sufficient evidence for a complete drag-only trajectory at any permitted wind. The geometry leaves the source of the lateral force unresolved; it does not uniquely identify spin.

In still air, each position bounds the slope of a line through the origin. For these positive coordinates, the closure of its admissible interval is

$$\left[\frac{y_i - \epsilon}{x_i + \epsilon},\, \frac{y_i + \epsilon}{x_i - \epsilon}\right].$$

The two intervals' closures first touch at $\epsilon = 7.279$ mm. Thus 5 mm precision resolves a departure from a straight horizontal path in this example. A 1 cm bound admits a straight path.

2 Aerodynamic models and conditional inference

2.1 Force laws and parameter ranges

Let $\boldsymbol{v} = \dot{\boldsymbol{r}}$ be the ground velocity, $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{w}$ the air-relative velocity, and $U = \|\boldsymbol{u}\|$. We propagate three force models,

$$\begin{aligned} \boldsymbol{a}_{\mathrm{D}} &= \boldsymbol{g} - \kappa U \boldsymbol{u}, \\ \boldsymbol{a}_{\mathrm{S}} &= \boldsymbol{a}_{\mathrm{D}} + \boldsymbol{\beta} \times \boldsymbol{u}, \\ \boldsymbol{a}_{\mathrm{E}} &= \boldsymbol{g} - KC_D U \boldsymbol{u} + KC_L U^2 \boldsymbol{n}. \end{aligned} \tag{3}$$

D represents quadratic drag. S adds a transverse force with constant coefficient vector $\boldsymbol{\beta}$. In E, $K = \rho\pi R^2/(2m)$ and $\boldsymbol{n}$ points along $\boldsymbol{\omega} \times \boldsymbol{u}$. Lift vanishes when this cross product is zero. We use $g = 9.81$ m/s^2 downward, $R = 0.11$ m, $m = 0.43$ kg and $\rho = 1.20$ kg/m^3. With all inputs in SI units, E uses

$$\begin{aligned} C_D &= 0.155 + \frac{0.346}{1 + \exp[(U_c - 12.19)/1.309]} + \delta, \qquad U_c = \min(29.9, \max(9.3, U)), \\ C_L &= 0.38 \tanh\!\left(5R\|\boldsymbol{\omega} \times \boldsymbol{u}\|/U^2\right). \end{aligned} \tag{4}$$

The drag transition comes from a wind-tunnel fit for a nonspinning Teamgeist ball [2]. We clip its speed input to the initial-speed range of a separate 29-flight study [3]. The offset δ and the saturating lift law are modelling assumptions. Clipping a curve does not validate it for the unknown ball or for speeds outside its measured domain. S is an effective transverse-force model: its $\boldsymbol{\beta}$ is not a measured angular velocity. The broader E spin prior is a scenario range rather than an inferred population distribution.

The prior probabilities of D, S and E are $0.2, 0.4, 0.4$. We draw constant horizontal wind uniformly from a disk of radius 3 m/s. Independent uniform launch-velocity ranges are $[18, 38]$, $[-12, 12]$ and $[1.5, 11]$ m/s along x, y, z. In D and S, κ is uniform on $[0.006, 0.024]$ m^{-1}. In S, each component of $\boldsymbol{\beta}$ is uniform within ± 0.8 s^{-1}. In E, each component of $\boldsymbol{\omega}$ is uniform within ± 65 rad/s and δ is uniform on $[-0.04, 0.27]$.

2.2 Why modest aerodynamic uncertainty matters

The first chord gives a speed scale $U_0 \simeq 32$ m/s. With $\kappa \simeq 0.020$ m^{-1}, the drag time scale $(\kappa U_0)^{-1} \simeq 1.56$ s exceeds the approximately 0.4 s flight, but their ratio is about 0.26 and cannot be neglected. A transverse coefficient of 0.5 s^{-1} gives an initial acceleration scale of 16 m/s^2. Over the remaining 0.30 s, even an unresolved 1 m/s^2 acceleration changes a constant-acceleration extrapolation by about 4.5 cm. These are dimensional estimates, not extra simulated results. They explain why fitting the first 100 ms tightly need not fix the endpoint.

With constant wind, the force models also obey an energy check. Since $\boldsymbol{u} \cdot (\boldsymbol{\beta} \times \boldsymbol{u}) = 0$ and $\boldsymbol{u} \cdot \boldsymbol{n} = 0$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{U^2}{2} = \boldsymbol{u} \cdot \boldsymbol{g} - d(U)U^2, \qquad d(U) \geq 0.$$

The transverse term redirects air-relative velocity; drag dissipates air-relative kinetic energy. This identity constrains the acceleration bounds used in the match-data check.

2.3 Trajectory weighting and goal-plane probabilities

After drawing wind and aerodynamic parameters, we centre a launch-velocity proposal on a gravity-based estimate corrected three times against the latest visible position. We draw velocities from a mixture of this local uniform proposal (half-width $0.007\,\mathrm{m}/t$ per axis at observation time t) and the original prior (2%). A trajectory inside every observed interval receives the prior-to-proposal density ratio. Others receive zero weight. More generally, with parameters θ, proposal density q_M and observation likelihood L,

$$w_{Mj} = \frac{p(\theta_{Mj} \mid M) L(D \mid \theta_{Mj}, M)}{q_M(\theta_{Mj})}, \quad \widehat{Z}_M = \frac{1}{N_M}\sum_j w_{Mj}, \quad \widehat{p}(M \mid D) \propto p(M)\widehat{Z}_M. \tag{5}$$

Nuisance parameters drawn from their priors cancel in the ratio; the velocity proposal still requires correction. The mean weight over all proposals estimates each model's evidence. Multiplying by the model priors and normalizing yields the posterior weights. For propagation to the goal, systematic resampling caps each model at 512 trajectories per test kick or 8,192 per seed for the official case. The official runs stayed below the cap.

We integrate by fourth-order Runge–Kutta with steps of 25 ms through the official observations and at most 12.5 ms thereafter. Linear interpolation locates and orders plane and ground events. Across four numerical checks, including independent DOP853 integrations in SciPy [4], discrepancies stayed below 0.171 mm in position and 0.038 ms in time. After summing weights by first event, we

distribute the direct-entry mass across a 30×10 equal-area grid. The centre of the highest-mass cell is the point forecast. Cells ranked by decreasing mass form the raw joint region once their cumulative mass reaches 90%. Zero total weight leaves the forecast undefined. Zero direct-entry weight leaves its conditional map undefined. We monitor ESS $= 1/\sum_j \bar{w}_j^2$ using normalized weights (and model/phase mixture contributions), flagging values below 25 for test kicks. ESS is a sampling diagnostic, not evidence of physical validity. A finite run with no accepted proposals is not a proof that the allowed parameter set is empty.

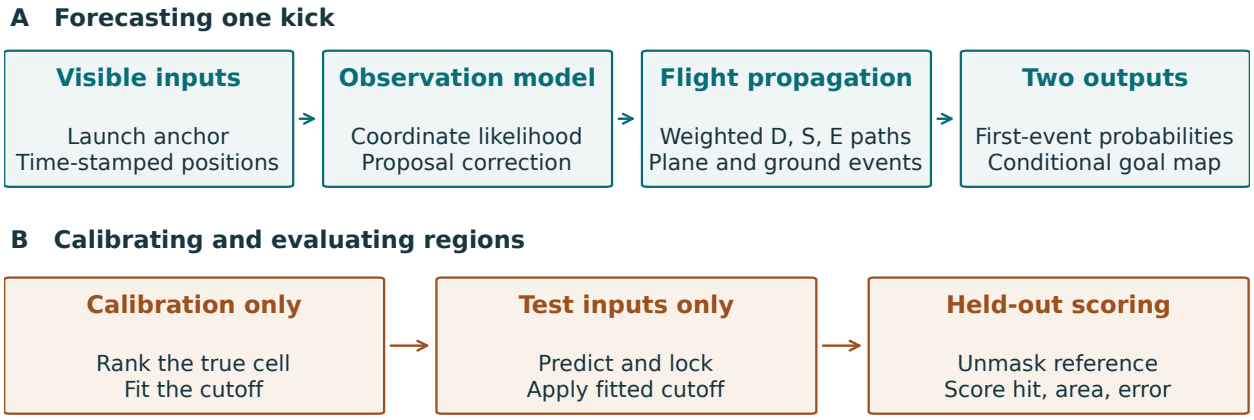


Figure 2. From positions to a scored region. A shows the single-kick prediction. First-event probabilities describe outcomes including ground contact, while the location map is conditional on direct entry. B shows the separate calibration and test operations in Sec. 6. Held-out reference positions enter only during scoring.

3 Goal-line forecast for the prescribed hypothetical kick

3.1 Modal cell and joint prediction region

The highest-probability cell is centred at approximately $(Y, Z) = (2.56, 2.07)$ m and holds 29.14% of the probability conditional on direct entry (Fig. 3). Its centre is 1.098 m in from the positive-y post and 0.366 m below the crossbar. The six most probable cells form the rectangle

$$2.196 \le Y \le 2.928 \text{ m}, \qquad 1.708 \le Z \le 2.196 \text{ m}. \tag{6}$$

Together they cover 0.3572 m^2 and contain 90.52% of the conditional probability. Equation (6) gives the edges of a joint two-dimensional region. The median time from release to crossing is 0.399 s, with 5th and 95th percentiles of 0.392 and 0.407 s.

We combined two independent runs of 131,072 proposals per model per seed. All 3,469 positive-weight trajectories crossed inside the opening, giving an estimated direct-entry probability of 1 in this sample. The posterior model weights were $0, 0.346, 0.654$ for D, S and E. The zero weight for D agrees with the geometric bound in Eq. (2). The prescribed hypothetical example supplies no measured crossing for comparison. The estimate of one is conditional on the adopted parameter support and finite sample; it is not a guarantee of scoring or a calibrated real-world goal probability.

3.2 Sensitivity to grid choice and force law

Both seeds selected the same cell, as did mixtures in which E accounted for 10% to 90% of the combined S/E prior. A finer grid moved the mode to $(2.501, 2.013)$ m, and shifting the original

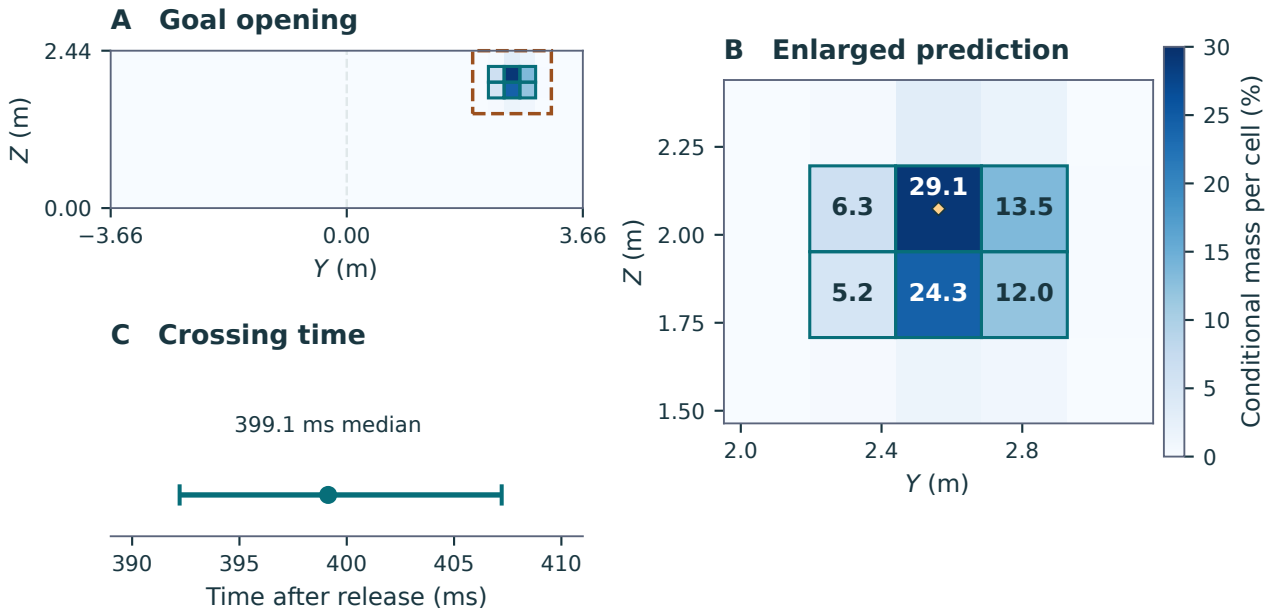


Figure 3. The prescribed kick concentrates in six upper-right cells. A locates the dashed enlargement in the full opening. B shows the conditional direct-entry probability per 0.244 m square cell on the same colour scale as A. The outlined cells form the raw 90% joint region. The diamond marks the modal centre. C shows the 5th, 50th and 95th percentiles of direct-entry crossing time. These are conditional predictions for the hypothetical kick.

grid by half a cell moved it to $(2.684, 1.952)$ m. The unrounded centre $(2.562, 2.074)$ m identifies the grid cell for reproduction. The six-cell region gives the scale of its positional uncertainty. The conditional continuous mean is $(2.603, 1.970)$ m; it minimizes posterior squared-distance loss, whereas the modal cell answers a fixed-area probability question. We retain the modal-cell convention for all scored forecasts. Neither estimator is a grid-independent continuous density mode, and the displayed millimetre digits do not represent millimetre accuracy.

Kinematic fits illustrate the effect of force assumptions over the remaining eight metres. A least-squares fit of launch velocity under gravity alone gives $(3.217, 1.957)$ m. Constant acceleration fitted exactly to release and both frames gives $(2.495, 1.920)$ m. This spread measures sensitivity to the extrapolation law.

4 A lower bound on two-position prediction error

4.1 Indistinguishable readings and distinct crossings

The prescribed readings at 50 and 100 ms admit two physically distinct crossings even when we fit the displayed coordinates to numerical precision. In model S we set the wind to zero and fixed β_x at either -0.5 or $+0.5$ s^{-1}. For each choice we solved the six early-coordinate equations for the three launch-velocity components and κ, β_y, β_z. Both solutions lie within the prior ranges. Their largest coordinate residual was below 5×10^{-13} m, and an independent integration with MATLAB `ode45` reproduced both crossings [5].

The two fitted drag coefficients agree to the displayed precision, 0.02007 m^{-1}. The corresponding (β_y, β_z) pairs are $(-0.206, -0.463)$ and $(0.085, -0.251)$ s^{-1}. At any instant, the component of $\boldsymbol{\beta}$ parallel to $\boldsymbol{u}$ contributes no transverse force. As gravity changes the direction of $\boldsymbol{u}$, that component

can begin to affect the flight. Small compensating changes to launch velocity allow the two force histories to match both early readings.

The crossings are $Q_A = (2.57378, 2.00251)$ m and $Q_B = (2.62831, 1.93396)$ m, separated by 8.76 cm. Both trajectories give exactly the same displayed centimetre readings. A point predictor receiving those readings must return the same $\widehat{Q}$ for both. By the triangle inequality, its error on at least one member of this pair satisfies

$$\max\{\|\widehat{Q} - Q_A\|, \|\widehat{Q} - Q_B\|\} \;\geq\; \tfrac{1}{2}\|Q_A - Q_B\| \;\approx\; 0.0438 \text{ m}. \tag{7}$$

This lower bound applies whenever both flights are admissible. It requires no assumption about their relative frequency.

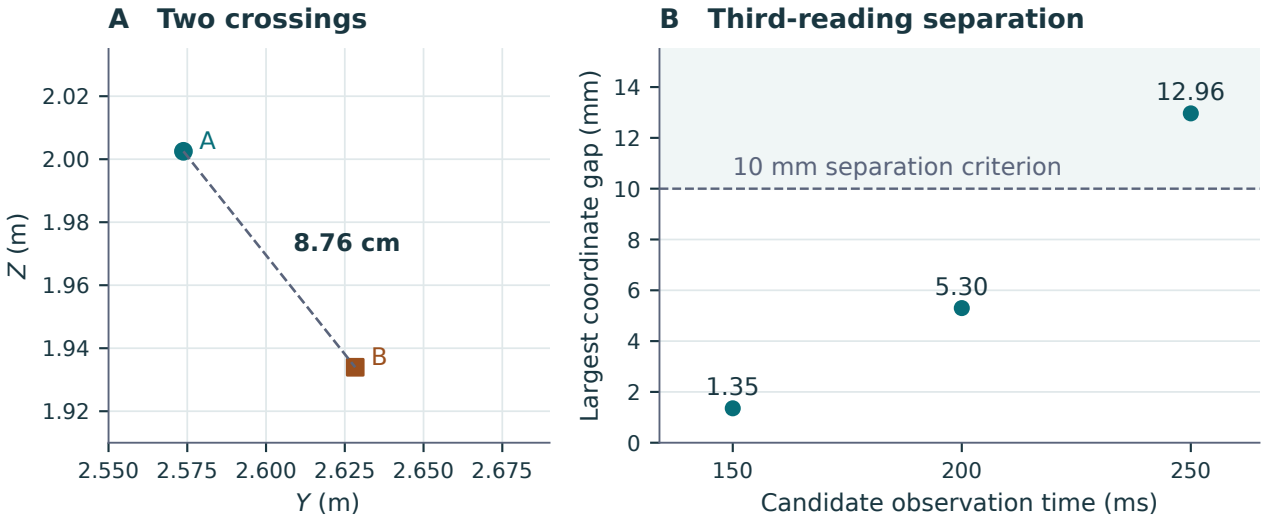


Figure 4. Identical early inputs can lead to different goal-plane crossings. A shows two admissible model-S endpoints at equal spatial scales. Their separation enters Eq. (7). B gives the largest coordinate gap at three candidate observation times. Under the assumed 5 mm bound per coordinate, the two possible reading boxes are disjoint when the gap exceeds 10 mm.

5 Observation uncertainty and sensitivity

5.1 A measurement layer separate from the force law

For coordinate axis a, let d_{ia} be a displayed position with quantum $q = 0.01$ m. A tracking model before rounding is

$$d_{ia} = \mathcal{R}_q[r_a(t_i + \tau; \theta, M) + b_a + e_{ia}], \quad b_a \sim N(0, \sigma_b^2), \quad e_{ia} \sim N(0, \sigma_e^2). \tag{8}$$

The bias b_a is shared within a kick; e_{ia} is independent across frames and axes. The release offset τ is shared across all frames. Conditional on b, τ, the likelihood is

$$L = \prod_{i,a} \left[\Phi\left(\frac{d_{ia} + q/2 - r_a(t_i + \tau) - b_a}{\sigma_e} \right) - \Phi\left(\frac{d_{ia} - q/2 - r_a(t_i + \tau) - b_a}{\sigma_e} \right) \right]. \tag{9}$$

Here Φ is the standard-normal cumulative distribution. The rounding-only case is the corresponding indicator likelihood. We sample b from its prior and integrate it through the importance weights. For bounded timing scenarios, τ is uniform on its stated interval and integrated with five or seven

Gauss–Legendre nodes, sharing a fixed total proposal budget. These likelihood parameters are known simulation inputs in the tests below; no field calibration is claimed.

The observation scale can rival the aerodynamic signal. If the release coordinate is exact and two frame errors have variance σ^2 and correlation ρ_e, the inferred discrete acceleration obeys

$$\operatorname{Var}\left(\frac{e_2 - 2e_1}{h^2}\right) = \frac{(5 - 4\rho_e)\sigma^2}{h^4}. \tag{10}$$

At $h = 50$ ms, a common 1 ms timing shift contributes $-v\tau/h^2$ for the linear motion term, or 12 m/s^2 in magnitude at 30 m/s. A translation shared by all three coordinates, including release, cancels in the second difference; a bias shared only by the two measured frames does not. This distinction prevents apparent curvature from being attributed automatically to lift.

5.2 Which assumptions move the prescribed prediction?

The nominal case and fourteen one-factor scenarios use the same two seeds and 131,072 proposals per model per seed. Table 2 and Fig. 5 compare continuous conditional means and nominal region areas after reconditioning on the same readings. Using the mean separates a displacement of the distribution from a grid-cell jump.

Table 2. Selected sensitivity scenarios. All other nominal conditions are held fixed. Shift is the Euclidean displacement of the continuous conditional mean; area is the nominal 90% grid region. Mass changes affect E's force-to-mass factor; the acceleration coefficients in D/S keep their definitions.

Changed assumption	Mean shift (cm)	Area (m^2)
Nominal assumptions	0.00	0.357
Coordinate half-width: 10 mm	14.11	0.893
Both observation times: +1 ms	3.73	0.357
Wind cap: 0 m/s	0.83	0.357
S: halve β component limits	3.97	0.357
E: halve ω component limits	9.08	0.357
E: ball mass 410 g	0.25	0.357
E: ball mass 450 g	0.45	0.357
6 mm independent + 8 mm shared noise	23.70	1.072

Doubling the coordinate bound to 10 mm moves the mean by 14.1 cm and increases the region to 0.893 m^2. Adding 6 mm independent tracking error and 8 mm shared bias before rounding moves it by 23.7 cm. Within these reconditioned scenarios, removing wind shifts the mean by only 0.83 cm, while halving E's spin range shifts it by 9.08 cm. The 410–450 g endpoints are consistent with the specified match-ball mass range [7]; radius is held fixed, and this is not a coupled ball-size study.

Tightening the bound to 2.5 mm produces ESS 68, below the prescribed-case threshold of 200. Shifting both observation times earlier by 1 ms leaves ESS 1; either sign of a 5 ms shift yields no positive-weight samples. These cases diagnose inadequate support or sampling, not precise endpoints. Interactions among timing, noise, wind and spin remain unresolved by one-factor tests. In particular, the small wind effect is local to this conditioned example and cannot be generalized to all penalties.

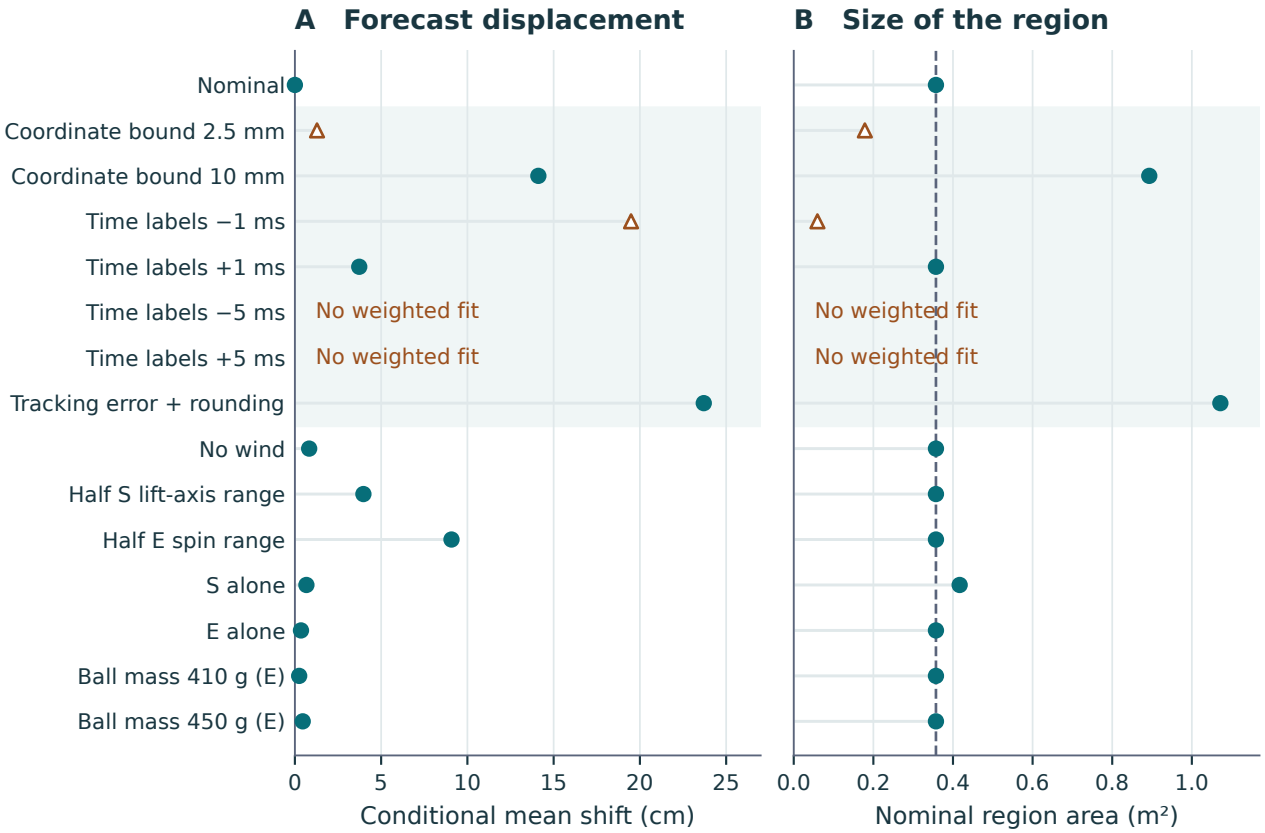


Figure 5. Observation assumptions move the forecast and change its size. A measures displacement from the nominal conditional mean. B shows the area of the nominal 90% region, with the baseline area dashed. Shading groups observation scenarios. Open triangles flag runs below the prescribed-case ESS threshold of 200. Rows with no positive weight have no plotted estimate.

6 Does the second reading help at 100 ms?

6.1 Fixed inputs, budgets and comparison targets

We generate 90 calibration kicks and 120 fresh test kicks at previously unused seeds. The force generator comprises balanced blocks of zero spin, sustained spin and low-spin wake trajectories. Its drag, Magnus and wake laws differ from those used for inference (Appendix D). The reference solver uses a 1 ms time step. We retain 55 and 71 direct entries for region calibration and test, respectively. Non-entry shots remain in the event-probability evaluation. Each trajectory is one statistical unit, with one inference seed per shot and method in each observation profile.

The rounded profile contains centimetre quantization. The tracking profile uses the same shots and adds two errors to the exact coordinates before quantization: independent 6 mm standard deviation per coordinate per frame, and a shared 8 mm standard deviation per coordinate for each kick. Both profiles have exact frame times and release position. The tracking inference uses the corresponding noise model, so its result is conditional on this assumed observation specification. We evaluate other timing and error interactions separately.

The late-frame and two-frame aerodynamic methods share the same model weights, parameter supports, solver and 8,192 proposals per model. Each model propagates up to 512 samples, with the actual count depending on positive-weight support. The methods share a 100 ms input cutoff, after which computation begins. The late-frame method receives the 100 ms position and the two-frame method receives both prescribed times. Each shot uses the same inference seed across methods and

profiles. The gravity control uses both readings. Its deterministic gate position is the centre of a distance-ranked region calibrated on the same shots. It predicts direct entry with a binary probability. The aerodynamic forecasts use their maximum-mass grid cells as point estimates. The comparison evaluates the complete procedures, including their distinct point conventions.

Calibration and test inputs are generated separately. The predictor reads only the input arrays. Its outputs are stored and hashed before computing scores against the corresponding references. The nominal 90% target, a common target grid from 50 to 100%, the three prediction methods and the family and boundary subgroups are fixed in the run configuration. Thresholds are learned once per method and profile using the calibration split.

6.2 Calibrated regions on the same goal grid

Let $H_D(j)$ be the conditional mass in goal cell j. Let $j_1, \dots, j_J$ rank cells by decreasing mass with a fixed tie order. For a cell with positive mass define the entry score

$$S_D(j_k) = \sum_{\ell<k} H_D(j_\ell). \tag{11}$$

Assign score one to empty cells, including all cells when the location map is absent. For gravity, let d_j be the Euclidean distance from the cell centre to the ballistic crossing. Set $S_D(j) = d_j/(L + d_j)$, where L is the goal diagonal. This monotone transform gives the same nested sets as a radius cutoff. An undefined ballistic crossing assigns score one to every cell.

For a coverage target c, sort the 55 scores of the true cells from the calibration entries and set q_c to order statistic $\lceil (55 + 1)c \rceil$. If the rank exceeds 55, set $q_c = 1$. We include every cell with $S_D(j) \le q_c$. At $q_c = 1$ the set is the whole opening. This fallback is visible in the mean area. The ranking device follows the split-calibration principle [8]. An exchangeable calibration and test population admits a rank argument for marginal coverage. Here, the generator uses stratified blocks and the real tracking process is unknown. We report empirical coverage in the declared test population rather than treating the target as a guarantee.

Table 3. Held-out comparison at a common 90% calibration target. Region hits, area and point error refer to 71 direct entries. Event Brier scores refer to all 120 kicks per method and profile. Gravity uses two observations and a calibrated radius.

Profile	Forecast	Hits	Area (m^2)	Mean error (m)	Brier
rounded	100 ms only	64/71	2.003	0.208	0.070
rounded	50 + 100 ms	67/71	0.381	0.170	0.018
rounded	Gravity + radius	70/71	1.222	0.186	0.083
tracking	100 ms only	62/71	1.825	0.216	0.075
tracking	50 + 100 ms	60/71	0.859	0.255	0.078
tracking	Gravity + radius	69/71	1.255	0.199	0.075

6.3 The gain depends on the measurement profile

For rounded inputs at the 90% target, both readings cover 67/71 crossings in a mean area of 0.381 m^2. The late frame covers 64/71 in 2.003 m^2. The 95% Wilson intervals are 86.4–97.8% and 81.0–95.1%, respectively. Equal targets permit different achieved coverage. At the separately calibrated 95% operating point, both methods cover exactly 70/71 in this test. The mean areas are 0.585 and 3.482 m^2, respectively. Gravity also covers 70/71 at its 90% target, with mean area 1.222 m^2. These



Figure 6. Region size at equal realized coverage depends on measurement error. A and B show held-out hits and mean area (log scale) for each independently calibrated target. Open markers indicate the 90% target. C and D give selected operating points with the same number of hits within each profile. The late-frame curve has no sampled point at 67/71. All panels use the same 71 synthetic direct entries. The whole goal opening is 17.86 m^2.

descriptive comparisons match achieved coverage among the prespecified operating points. Methods can miss different kicks at the same achieved coverage.

The two-frame point error is lower than the late-frame error by 3.80 cm on average. A paired 4,000-resample shot bootstrap gives a 95% interval of 0.22–7.62 cm for the reduction. The corresponding interval for a reduction relative to gravity is −2.29 to 5.85 cm. Gravity remains competitive for point estimation. The two-frame event Brier score is 0.018 against 0.070 for the late frame and 0.083 for the binary gravity forecast. A small region and a good point are different achievements.

With tracking noise, the 90% target gives 60/71 hits for two frames, 62/71 for the late frame and 69/71 for gravity. The two-frame area is smaller at this target, with lower observed coverage. At equal observed coverage of 67/71, the two-frame procedure at its 95% target uses 1.385 m^2. The gravity radius at its 80% target uses 0.933 m^2. These different calibration targets identify two operating points of equal test coverage. They do not imply that the two methods achieve the same coverage on future data. The two-frame mean point error exceeds the gravity error by 5.57 cm, with a paired 95% interval of 1.17–10.25 cm. The intervals for the noisy two-frame versus late-frame point and event differences include zero.

The area ordering of the two-frame model and gravity differs between these observed operating points. At $h = 50$ ms the curvature estimator magnifies millimetre-scale errors by h^{-2}. A shared spatial bias persists in the second difference because the release coordinate is fixed separately. Even with its assumed likelihood supplied, the two-frame procedure must infer flight parameters from a noisier curvature signal. The gravity model keeps a fixed acceleration and avoids estimating transverse acceleration from those differences. Its relative advantage on the noisy test is consistent with that reduction in estimation variance. These tests use prescribed tracking errors.

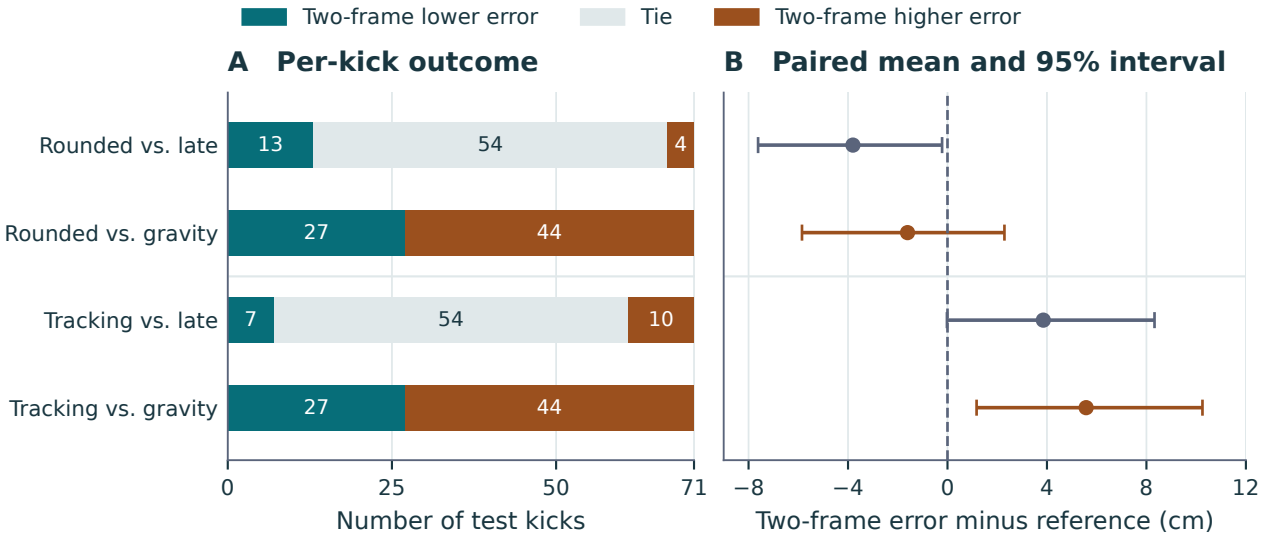


Figure 7. The point-error comparison changes when tracking error is added. A counts the direct-entry kicks on which two-frame aerodynamics gives a smaller, equal or larger Euclidean point error than the indicated reference. B gives two-frame error minus reference error and its paired 95% shot-bootstrap interval. Negative differences favour the two-frame aerodynamic estimate. Each row contains the same 71 kicks.

6.4 Availability and computational cost

All 71 direct entries have a defined point under all three procedures in both profiles. All 120 event probabilities per method are defined. One noisy late-frame forecast falls below the diagnostic threshold of 25 effective samples. Region thresholds near one can still make a result too broad to guide action even when the sampler returns a valid point. For example, the noisy two-frame set reaches full-goal area at the 97.5% calibration target. The tested distribution includes shots near the goal boundary. Appendix B reports their small denominators separately.

In the local Python run, the rounded two-frame prediction took 84 ms at the median and 94 ms at the 95th percentile. The noisy two-frame prediction took 121 and 131 ms. These figures include inference and gate-map construction on one host and exclude image capture, reconstruction, data transfer and movement. For an illustration of scale, 131 ms is about 44% of the nominal 299 ms remaining after the last frame. Actual time available for action depends on each kick's crossing time and all processing stages. The timed runs use the test budget.

7 Applicability to an early forecast

A practically useful result needs a region and an estimate of the time available to act. The nominal median of 0.399 s leaves 299 ms from the last prescribed frame. On a separate synthetic study, moving

the pair to 100 and 150 ms reduces mean point error from 0.182 to 0.126 m, with 50 ms less remaining flight time. A third frame at the same later deadline has more low-ESS results. Appendix E separates the cost of waiting from sampling failure. These comparisons concern crossing forecasts. Estimating save probability requires a goalkeeper model with measured position, movement onset and reach.

The three public penalties sampled at 25 Hz have later crossing references but uncertain release timing. Their early tracks cannot satisfy all of the launch and error assumptions of the nominal model. One late-frame estimate can be scored retrospectively. These cases are tests of input compatibility, with zero eligible release-synchronized cases under our independent-timing criterion. The data and force bounds are given in the field appendix.

The nominal ball-centre test is defined by the problem's opening. Frame collisions, rebounds and interventions would require additional states and event surfaces. They could change classifications near the edges. Aerodynamic coefficients and prior weights are scenario choices, and real kick populations can differ from the synthetic generator. The equal-area grid defines the modal-cell answer, and changing its resolution or origin can move the reported point. Report the joint region together with its coordinate extent and the conditions of the estimate.

Conclusion

At 100 ms the nominal most-probable cell is centred at approximately $(2.56, 2.07)$ m. The joint direct-entry region extends from 2.196 to 2.928 m in Y and 1.708 to 2.196 m in Z. Its 0.3572 m^2 holds 90.52% of the nominal conditional model mass, with a median arrival of 0.399 s. The witness pair limits the attainable precision of any single point in the allowed force family.

The independently calibrated 100 ms test measures the contribution of the first frame. Under centimetre rounding alone, the two-frame region is substantially smaller than the late-frame region at equal observed coverage of 70/71. Under the specified tracking noise the gravity radius achieves a smaller area at equal observed coverage of 67/71. The results make observation quality central to model choice. Numerical agreement and synthetic calibration support the conditional forecasts. Match data with independently verified release, calibration and goal-plane outcomes are needed to measure field performance.

A Timing and coordinate errors at the crossing event

A.1 Closed-form control and event projection

With a fixed launch position and readings at h and $2h$, the constant-acceleration interpolant uses

$$\widehat{\boldsymbol{a}} = \frac{\boldsymbol{r}_2 - 2\boldsymbol{r}_1 + \boldsymbol{r}_0}{h^2}, \qquad \widehat{\boldsymbol{v}}_0 = \frac{\boldsymbol{r}_1 - \boldsymbol{r}_0}{h} - \frac{h}{2}\widehat{\boldsymbol{a}}. \tag{12}$$

For the prescribed readings this yields $(30.4, 9.4, 6.9)$ m/s and $(-16, -16, -12)$ m/s^2. This is a kinematic diagnostic. It assumes constant acceleration beyond the observations. Write $X = 11 - x_0$, $D = v_{0x}^2 + 2a_x X$, and assume a positive, transversal first forward root. A numerically stable evaluation is

$$T = \frac{2X}{v_{0x} + \sqrt{D}}, \quad \dot{x}(T) = \sqrt{D}. \tag{13}$$

The nominal values are $T = 0.4050$ s and $(Y, Z) = (2.495, 1.920)$ m. Equation (6) is obtained from the aerodynamic inference rather than this control.

Suppose an input perturbation changes the interpolated curve at fixed T by $\delta \boldsymbol{r}(T)$. Differentiating the event condition $x(T) = 11$ gives

$$\delta T = -\frac{\delta x(T)}{\dot{x}(T)}, \qquad \delta Q = P\,\delta \boldsymbol{r}(T), \qquad P = \begin{pmatrix} -\dot{y}(T)/\dot{x}(T) & 1 & 0 \\ -\dot{z}(T)/\dot{x}(T) & 0 & 1 \end{pmatrix}. \tag{14}$$

This expression converts longitudinal errors into lateral and vertical errors by the local slope of the path. A perturbation parallel to the local velocity changes the time to first order and cancels out of Q. Near $\dot{x}(T) = 0$ the derivative can grow without bound, and when the forward root disappears the linearized location is undefined.

Let $s = T/h$, $\lambda_1 = s(2 - s)$, and $\lambda_2 = s(s - 1)/2$. These are the Lagrange weights of the two observed positions when the launch anchor is fixed. With spatial errors $\boldsymbol{e}_i$ and a shared true time shift τ relative to the time labels, the first-order event shift is

$$\delta Q = P\left[\lambda_1 \boldsymbol{e}_1 + \lambda_2 \boldsymbol{e}_2 + \left(\lambda_1 \boldsymbol{v}(h) + \lambda_2 \boldsymbol{v}(2h)\right)\tau\right]. \tag{15}$$

At the nominal root the weights are -49.41 and 28.76. Independent 5 mm per-axis bounds contribute first-order shifts of at most 0.439 m in Y and 0.424 m in Z. A common 1 ms offset contributes $(-0.123, -0.093)$ m for positive offset under the stated sign convention. Central differences at $1\,\mu$s reproduce the timing derivative to within 2×10^{-5} m/s. The large values characterize extrapolation by an unconstrained quadratic. The main model restricts curvature through force and launch ranges.

A.2 Joint perturbations and loss of a forward root

For each timing bound and spatial bound, evaluate both phase endpoints and zero phase, and the corners of the six-coordinate box. A zero bound uses one value rather than repeating identical inputs. Refit the polynomial and solve the forward plane event for each case. Ground and goal-edge events are disabled in this diagnostic to isolate the existence of a forward plane root. The maximum shift is taken over tested roots only. A nonlinear map can attain its extrema away from the sampled corners, so this maximum is a scenario statistic.

Table 4. Joint timing and position stress test for the quadratic control. First-order maxima are coordinatewise sums of absolute derivative contributions. The sampled maximum is a Euclidean shift at a forward plane root. Counts include all selected timing and spatial combinations.

Phase bound (ms)	Spatial bound (mm)	First-order Y (m)	First-order Z (m)	Sampled shift (m)	No root (count)
0	0	0.000	0.000	0.000	0/1
0	2.5	0.219	0.212	0.319	0/64
0	5	0.439	0.424	0.671	0/64
0	10	0.877	0.848	1.502	0/64
1	0	0.123	0.093	0.197	0/3
1	2.5	0.342	0.305	0.533	0/192
1	5	0.562	0.517	0.930	0/192
1	10	1.000	0.942	2.025	0/192
3	0	0.370	0.280	0.297	1/3
3	2.5	0.589	0.492	0.537	64/192
3	5	0.808	0.704	2.298	48/192
3	10	1.247	1.128	4.487	32/192

The quadratic can lose its plane crossing at a 3 ms timing bound. At 1 ms and 5 mm, all 192 combinations cross and the largest sampled displacement is 0.930 m. The corresponding coordinatewise

linear bounds are 0.562 and 0.517 m. Larger perturbations change the root substantially, making the local derivative a poor global description. The different failure counts across spatial bounds reflect the changing discrete corners, not a monotone probability of failure. This is a reason to carry event uncertainty through the solver when propagating input uncertainty. A fixed-time error band alone cannot represent event loss.

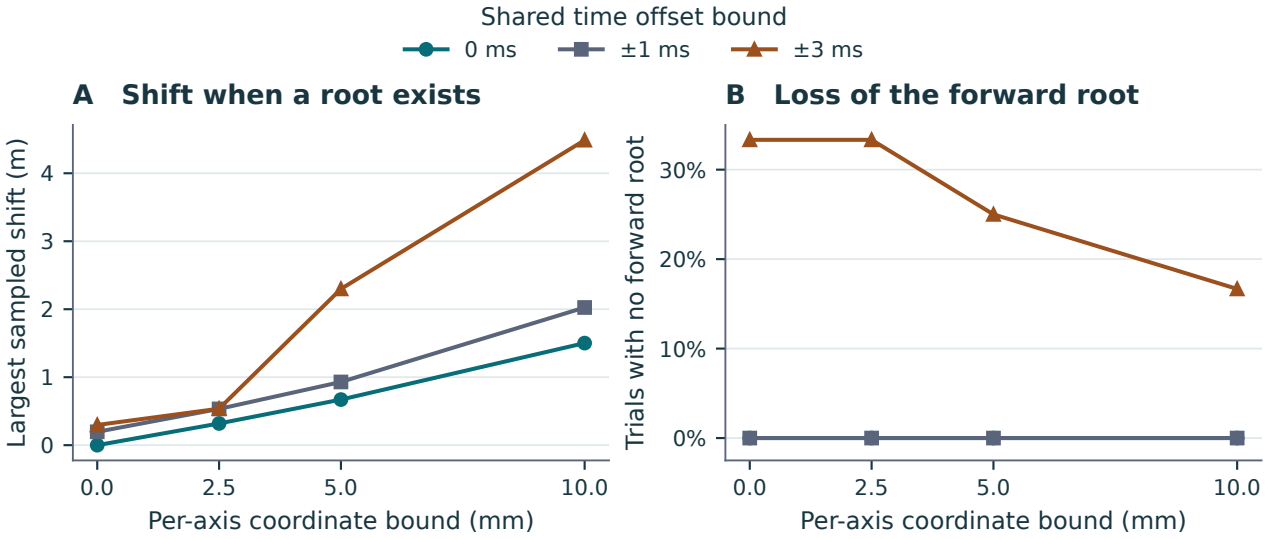


Figure 8. Input perturbations can move or remove the forward crossing. A measures the maximum displacement among sampled corners with a root in the quadratic control. B reports the fraction of enumerated combinations without a root. These are deterministic trial fractions for the stated timing endpoints and position corners. The diagnostic considers the plane alone.

A.3 Covariance propagation for correlated measurements

Under independent coordinate noise with covariance Σ_e in each frame, a shared bias covariance Σ_b, and an independent phase variance σ_τ^2, write $w = \lambda_1 \boldsymbol{v}(h) + \lambda_2 \boldsymbol{v}(2h)$. The first-order covariance is

$$\operatorname{Cov}(\delta Q) = P \left[(\lambda_1^2 + \lambda_2^2)\Sigma_e + (\lambda_1 + \lambda_2)^2 \Sigma_b + \sigma_\tau^2 w w^\top \right] P^\top. \tag{16}$$

The timing contribution is rank one. Its direction depends on the trajectory and event geometry. If a common coordinate translation were applied to release and frames alike, its effect would differ because the launch position would also change. The model uses an independently specified launch position. This formula separates local measurement amplification from model discrepancy and Monte Carlo variability. It is valid for a smooth forward crossing away from event switches.

B Operating points, subgroups and timing records

The operating points below report sample counts instead of treating calibration targets as test guarantees. Mean area includes every true direct entry. At a target of one the entire rectangular opening is returned by construction.

Near-edge means the minimum distance from the true direct entry to any of the four edges is below 0.244 m, the nominal cell width. The strata expose whether a successful overall average masks failures near the physical decision boundary. Family is available for evaluation only. The predictor never receives the synthetic truth label.

Table 5. Selected calibration targets and realized results from the 120-kick holdout. Rounded and tracking profiles share the 71 true direct entries.

Profile	Forecast	Target (%)	Hits	Mean area (m^2)
rounded	100 ms only	80	53/71	1.367
rounded	100 ms only	90	64/71	2.003
rounded	100 ms only	95	70/71	3.482
rounded	100 ms only	100	71/71	17.861
rounded	50 + 100 ms	80	58/71	0.187
rounded	50 + 100 ms	90	67/71	0.381
rounded	50 + 100 ms	95	70/71	0.585
rounded	50 + 100 ms	100	71/71	17.861
rounded	Gravity + radius	80	68/71	0.917
rounded	Gravity + radius	90	70/71	1.222
rounded	Gravity + radius	95	71/71	2.296
rounded	Gravity + radius	100	71/71	17.861
tracking	100 ms only	80	59/71	1.203
tracking	100 ms only	90	62/71	1.825
tracking	100 ms only	95	69/71	2.569
tracking	100 ms only	100	71/71	17.861
tracking	50 + 100 ms	80	54/71	0.523
tracking	50 + 100 ms	90	60/71	0.859
tracking	50 + 100 ms	95	67/71	1.385
tracking	50 + 100 ms	100	71/71	17.861
tracking	Gravity + radius	80	67/71	0.933
tracking	Gravity + radius	90	69/71	1.255
tracking	Gravity + radius	95	71/71	2.527
tracking	Gravity + radius	100	71/71	17.861

Table 6. Hits at the 90% calibration target by generator family and distance to the opening edge. The two edge strata partition direct entries. The three family strata form a separate partition. They are descriptive and were not separately calibrated.

Profile	Stratum	Entries	100 ms	50 + 100 ms	Gravity
rounded	no spin	24	24/24	24/24	24/24
rounded	stable spin	23	16/23	21/23	22/23
rounded	wake	24	24/24	22/24	24/24
rounded	near edge	11	9/11	10/11	11/11
rounded	interior	60	55/60	57/60	59/60
tracking	no spin	24	24/24	24/24	24/24
tracking	stable spin	23	14/23	12/23	21/23
tracking	wake	24	24/24	24/24	24/24
tracking	near edge	11	9/11	7/11	10/11
tracking	interior	60	53/60	53/60	59/60

Table 7. Numerical availability and wall-clock timing in a single local run. Runtime includes inference and construction of the predicted region ranking inputs. The ballistic timing covers its trajectory result. Threshold application is omitted in both. Times are descriptive for this host and software environment.

Profile	Forecast	Points	Events	ESS < 25	Median (ms)	95th (ms)
rounded	100 ms only	71/71	120/120	0	90.9	103.1
rounded	50 + 100 ms	71/71	120/120	0	83.6	93.7
rounded	Gravity + radius	71/71	120/120	0	0.2	0.3
tracking	100 ms only	71/71	120/120	1	107.5	117.4
tracking	50 + 100 ms	71/71	120/120	0	121.4	130.7
tracking	Gravity + radius	71/71	120/120	0	0.2	0.3

The benchmark has one seed for each shot and profile. Paired intervals use 4,000 resamples of differences matched by shot ID. Event intervals use all shots. Position, area and hit intervals use the true direct entries. The test is balanced by generator family before observing outcomes. Intervals are descriptive of the sampled mixture. An independent field calibration would need known release, independent reference crossings, and an explicit protocol for intervening contacts.

C Implementation and physical checks

C.1 A callable forecast contract

The general method accepts the release point, a list of observed positions and timestamps, a camera-error model, and documented parameter supports. Input indices must be valid and coordinates finite. The solver advances each hypothesis from release to the last observation, rejecting a path that contacts the ground before the observations. Integrate surviving hypotheses until the earliest goal-plane crossing, descending ground contact, or horizon. The goal-plane coordinates are attached to an entry only when they lie in the open goal rectangle.

1. For each force family, sample wind and spin parameters. Form a launch-velocity proposal around the inversion of the last observation. Keep a 2% prior draw to provide support away from the local proposal.

2. Evaluate the observation likelihood at all supplied times and assign importance weights as in Eq. (5). If including a timing nuisance, distribute the fixed budget over quadrature nodes and combine their masses with the node weights.

3. Carry posterior mass to first events. For each crossing through the goal, assign its weight to a fixed cell. Normalize the event masses using the total weight and normalize goal-cell masses using the entry weight.

4. Return the event vector, modal cell, raw region and effective sample size. Report an unavailable event distribution when total mass is zero. If total mass is positive but entry mass is zero, report the event distribution and mark the conditional point and map undefined.

This separation avoids conflating a missed goal with an unsupported forecast. A boundary case can have a computed event vector and very little entry mass, so a conditional point should be accompanied by the unconditional entry probability. The physical support test and the numerical ESS warning answer separate questions.

Table 8. Roles of inputs in the nominal prediction. Changes to scenario choices require a fresh forecast.

Quantity	Role	Interpretation
Release, frame times, and opening	Given by the problem	Geometry and information budget
Ball mass, radius and air density	Nominal physical inputs	Set acceleration through the area-to-mass factor
Empirical drag curve	Literature-guided law	Limited range, with clipping and an offset prior
Wind and lift/spin bounds	Scenario and prior choices	Control admissible continuations and model weights
5 mm coordinate interval	Nominal rounding assumption	Relates reported positions to continuous paths
Cell resolution and RK4 steps	Numerical choices	Check stability before quoting modal coordinates
6 mm noise and 8 mm bias	Synthetic test parameters	Specify a noisy measurement regime

C.2 Model assumptions and checks

In the no-drag, no-lift limit, $x(t) = v_{0x}t$, $y(t) = v_{0y}t$, and $z(t) = R + v_{0z}t - gt^2/2$. This gives closed-form plane and ground times and checks event ordering in the integrator. Under still-air drag without lift, both horizontal acceleration components are proportional to the corresponding velocity components, so the horizontal heading remains fixed. The constant-wind criterion in Eq. (2) generalizes this relation in the wind frame. These limits check geometry and force signs independently of the posterior sampling.

For numerical cross-checks, the existing RK4 comparisons against adaptive DOP853 cover vacuum, drag with ground contact, effective lift and the speed-dependent empirical law. The largest reported event-position difference is 0.171 mm and the largest event-time difference is 0.038 ms. These four checks establish local numerical agreement under specified force laws. A finer solver cannot identify unknown wind, spin, or release-time error. The witness pair uses its own high-accuracy integration and a separate MATLAB replay.

D Synthetic generator and earlier comparisons

D.1 Synthetic truth and controls

The reference generator uses a different drag curve from the predictor,

$$C_D^{\mathrm{ref}} = 0.19 + \frac{0.20}{1 + \exp[(U - 20)/2.5]} + d_0, \qquad d_0 \sim \mathcal{U}(0, 0.1),$$

and a lift coefficient $0.32 \tanh(4R\|\boldsymbol{\omega} \times \boldsymbol{u}\|/U^2)e^{-0.12t}$, directed along $\boldsymbol{\omega} \times \boldsymbol{u}$. Launch velocities are independent uniforms on $[20, 36]$, $[-11, 11]$, $[2, 10]$ m/s, and wind is uniform in a horizontal disk of radius 2.5 m/s. Three equally represented families have spin-component bounds of 0, 55 and 6 rad/s. The last adds a time-varying transverse acceleration: project $A(0, \sin\phi(t), 0.4\cos\phi(t))$ perpendicular to $\boldsymbol{u}$, with $A \in [2, 5]$ m/s^2, period 0.18–0.55 s and uniform phase. Unsteady low-spin forces have experimental precedent [6], but these amplitudes, frequencies and family proportions are synthetic choices.

The original study splits whole kicks into 9 development, 36 calibration and 48 blind-test cases, generating reference flights with DOP853 [4]. Later extensions use new calibration/test seeds with the same physical generator, integrated by 1 ms RK4. Against the 48 frozen DOP853 flights, all

first-event labels agree; maximum endpoint and time differences are 1.9×10^{-6} m and 6.3×10^{-7} s. This checks numerical implementation, not the realism of the generator. Every forecast reads only its permitted early frames and is written and hashed before endpoint scoring. New-seed tests are conditional synthetic holdouts; they are not independent measurements of nature.

Point error is Euclidean distance on the goal plane. We report region hits and area on the same direct-entry population, and the Brier score $(P(G) - \mathbf{1}_G)^2$ for first-event classification. Gravity with a least-squares launch velocity and an exact constant-acceleration fit provide transparent controls. Missing event estimates and missing conditional points are reported separately. A smaller region is useful only with adequate coverage and forecast availability.

D.2 The second frame under equal proposal budgets

The original 48-kick blind study contains 33 direct entries, ten outside crossings and five early ground contacts. Its two-frame median point error was 0.133 m and raw-region coverage 30/33, but it used eight times the proposal budget of the single-frame baseline. We therefore base the information comparison on a retrospective rerun of those same kicks with 2,048 proposals per model for each input choice and three paired seeds (Table 9). The rerun is not a fresh blind test.

Table 9. Equal-budget retrospective comparison at the same 100 ms cutoff. The 99 goal predictions are 33 kicks times three seeds; they are not 99 independent kicks. Neither method lacks an event estimate in 144 attempts. Regions are uncalibrated.

Observed frames	Mean error (m)	Hits	Area (m^2)	ESS < 25
100 ms only	0.215	99/99	3.071	0
50 and 100 ms	0.170	91/99	0.288	11

The two-frame regions are much smaller, but their coverage also falls. Averaging seed differences within each kick and resampling 33 kicks 2,000 times gives a mean point-error difference (both minus late) of -4.46 cm, with a descriptive 95% bootstrap interval $[-9.14, 0.08]$ cm. The interval contains zero. The second frame supplies curvature information and reduces admissible trajectories, but these data do not establish a reliable mean point-error improvement. Sampling effort, coverage and area must accompany any accuracy claim.

D.3 Tracking error can defeat a narrow likelihood

A new 36-kick calibration set and 48-kick test set compare likelihoods using identical noisy inputs. Before centimetre rounding, each coordinate receives a kick-shared Gaussian bias of standard deviation 8 mm and frame-independent noise of standard deviation 6 mm. Every aerodynamic method uses 8,192 proposals per model and two paired seeds. The independent-error comparator has marginal standard deviation 10 mm; the shared-error model uses the actual simulated 6/8 mm decomposition. Their local velocity-proposal half-width is increased to $0.026\,\mathrm{m}/t$, while the rounding-only comparator retains $0.007\,\mathrm{m}/t$; importance weights correct each proposal. This compares complete finite-budget configurations, not likelihood changes under an identical proposal. It tests a known-noise scenario rather than estimating those parameters from two observations.

For each method and seed, calibration ranks goal cells by decreasing probability and records the cumulative probability preceding the true cell. A true cell with zero mass receives score one. With n direct entries in the separate calibration set, the cutoff is order statistic $\lceil 0.9(n+1) \rceil$; an unavailable rank or a cutoff of one returns the whole goal. Cells with preceding mass at most the cutoff are included. This finite-sample construction is useful under exchangeable calibration/test conditions;

Table 10. Noisy-input tests at 100 ms. There are 36 true direct entries and two seeds (72 attempts). n_Q counts defined point forecasts. Region hits and mean areas include all 72 attempts, assigning an empty raw region when the conditional map is undefined. Calibration may fall back to the entire opening.

Likelihood	Mean error (m)	n_Q	Raw hits	Raw area (m^2)	Calibrated hits	Cal. area (m^2)
Rounding only	0.425	68	27/72	0.355	72/72	17.861
Independent error	0.231	72	72/72	1.870	68/72	1.261
Shared + independent	0.257	72	71/72	1.256	71/72	1.609

the balanced synthetic families and real-world distribution shift preclude a blanket field-coverage guarantee. We report realized counts.

The rounding-only method attains 72/72 calibrated hits only by selecting the full 17.861 m^2 goal. Allowing tracking error yields materially more informative regions. Shared-noise modelling reduces mean paired error relative to the hard likelihood by 15.86 cm, with a shot-bootstrap interval of 4.95–26.53 cm on the 34 complete paired kicks. Its advantage over independent noise is not established: the paired interval includes zero. Gravity alone has mean point error 0.220 m, below both probabilistic noise models, but a worse event Brier score (0.104 versus 0.069 and 0.064). The rounding-only Brier score is 0.101 on 94/96 available estimates; the others use 96/96. Thus no single method dominates every metric, and successful noise modelling must not be described as universal point-accuracy superiority.

E When is another observation worth waiting for?

E.1 Later information versus more information at one deadline

Using the same new synthetic calibration/test kicks, we compare three noiseless-rounded input choices with 8,192 proposals per model and two paired seeds. The original 50/100 ms pair is available at 100 ms. The 100/150 ms pair and all three positions are available at 150 ms. The former comparison includes waiting; the latter holds the decision time fixed.

Table 11. Additional-observation tests. The 72 goal attempts are 36 kicks times two seeds; event availability and low ESS use all 96 attempts. n_Q is the number of defined goal-point forecasts. Empty maps count as misses and zero area, so missingness must be read alongside apparent area reductions.

Frames (ms)	Mean error (m)	n_Q	Hits	Area (m^2)	No event estimate	ESS < 25
50, 100	0.182	72	61/72	0.346	0	0
100, 150	0.126	72	62/72	0.136	0	20
50, 100, 150	0.130	65	52/72	0.093	6	46

Waiting for the later pair lowers mean error by 5.65 cm (36 complete paired kicks; 95% shot-bootstrap interval 2.79–8.98 cm), while mean remaining flight time falls from 321 to 271 ms. Calibrated hits increase from 61/72 to 64/72, with mean area decreasing from 0.350 to 0.177 m^2. This is evidence for a conditional forecast improvement at a known 50 ms time cost, not evidence for an increased save probability.

At the same 150 ms cutoff, retaining the earliest frame produces six unavailable event estimates and low ESS in 46/96 attempts. Only 31 kicks have defined paired points for both seeds. On those complete pairs, three frames minus the late pair changes mean error by +0.27 cm, with interval

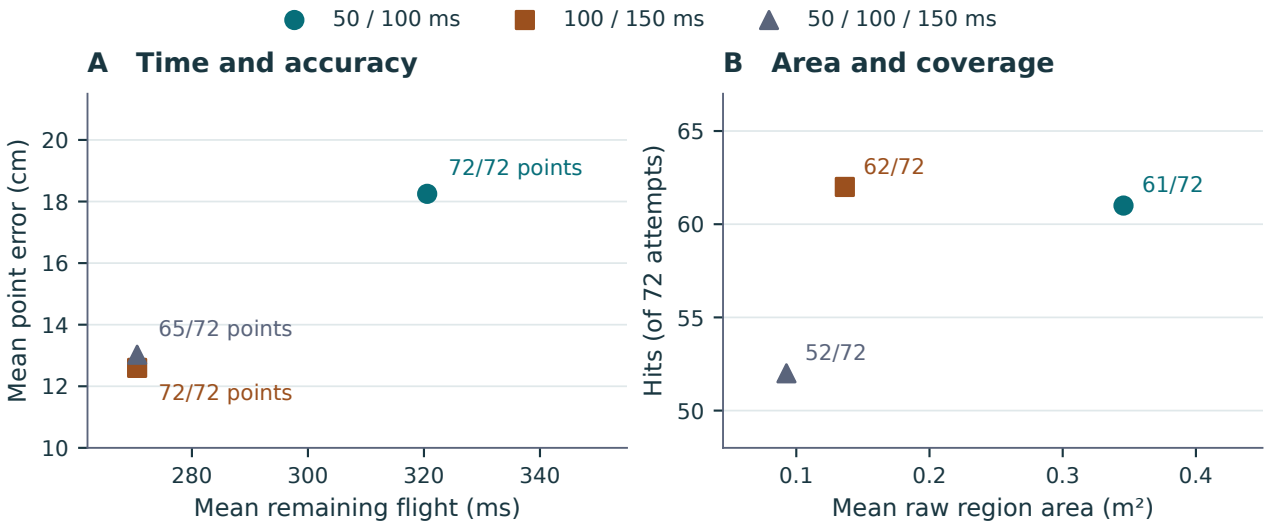


Figure 9. Waiting trades 50 ms for a more accurate pair of readings. In A, point errors use defined goal forecasts and remaining time averages all 36 true entries. The three-frame mean uses 65/72 available estimates. B displays raw region area and coverage together. Undefined maps count as empty, as in Table 11. The simulated time window excludes computation and goalkeeper motion.

$[-1.71, 2.28]$ cm. The small three-frame region accompanies reduced raw coverage (52/72), so it is not an improvement by itself. More observations do not guarantee lower realized prediction error, even with exact inference. Here, low ESS and missing estimates identify finite-proposal concentration as a contributor; model mismatch and finite-test variation may also matter. This does not establish that the third observation is intrinsically harmful. An adaptive proposal or sequential sampler is a computational extension requiring fresh validation, not a result established here.

E.2 Distinguishing the flights with a third observation

At 150, 200 and 250 ms the largest coordinate differences are 1.35, 5.30 and 12.96 mm, respectively (Fig. 4B). With deterministic centimetre rounding, the heights already display as 1.26 and 1.25 m at 200 ms. Under a general ±5 mm error bound, either flight can still yield the same reading at that time. Of the three times examined, only 250 ms separates this pair for every admissible error.

E.3 Frames must arrive before the decision

A separate set of 48 synthetic kicks (16 per force family, 33 direct entries) tests acquisition under 13 camera/noise profiles. For each deadline, 50 and 100 ms, we choose the latest distinct native frame already received and no more than 10 ms old, and use its native camera timestamp relative to estimated release in inference. With 2 ms frame availability delay, 25, 60, 120 and 240 Hz supply both frames for 0, 22, 47 and 48 kicks, respectively. These counts depend on simulated camera phase and the stated age criterion; they are not general hardware specifications. At 240 Hz, increasing delay to 6 ms lowers readiness to 46/48. Predictor execution time is additional.

For regular frames without drops, the worst-case age is bounded by $\ell + 1/f$, where ℓ is availability delay and f is frame rate. The 10 ms target with 2 ms delay therefore requires at least 125 Hz in this ideal periodic model. A 240 Hz pilot provides margin, but its frame interval alone brackets release only to a half-width of about 2.08 ms. A proposed 1 ms timing target needs a finer independent timing reference.

For the same simulated ±3 ms release error, marginalizing that offset increases ESS-qualified

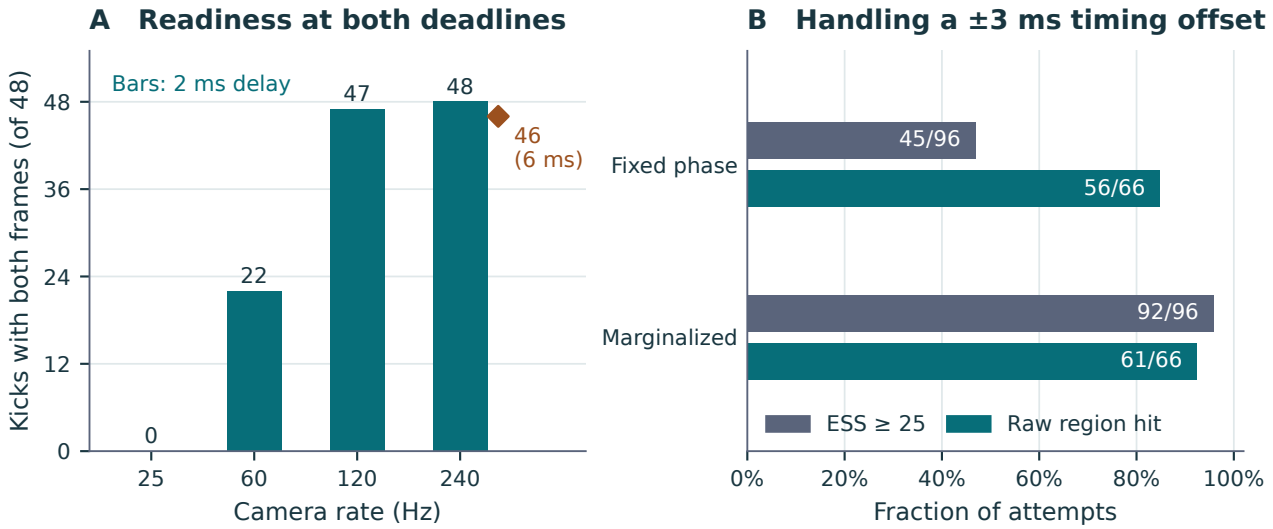


Figure 10. Frame availability and timing treatment affect different stages. A counts the 48 synthetic kicks with distinct timely frames at both deadlines. B compares a fixed release time with marginalization under the same simulated ±3 ms release offset. ESS uses all 96 attempts (48 kicks and two seeds). Raw region hits use the 66 direct-entry attempts.

estimates from 45/96 to 92/96 and raw hits from 56/66 to 61/66. Mean point error rises from 0.196 to 0.218 m and mean area from 0.377 to 0.543 m^2. Wider, better-supported uncertainty can be appropriate even when the point estimate does not improve. These capture simulations specify trial targets; they do not replace an actual calibration or field measurement.

F Compatibility with 25 Hz match tracking

F.1 Match timing and reference crossings

The seven matches in the IDSSE v1 elite-soccer dataset contained three executed penalties, all recorded as goals [9]. We translated the longitudinal coordinate by 41.50 m into the penalty-spot frame and took the first live frame after a dead-ball interval as an exact anchor. The next two tracking frames fell at +40 and +80 ms. Both input choices were evaluated at +80 ms with the original priors and 5 mm coordinate bound. We interpolated each reference crossing between later frames from the same system, at least 0.32 s after the forecast cutoff. We isolated inputs by time and locked predictions before scoring. Prior knowledge of these crossings from data screening makes the comparison retrospective.

Table 12. Three penalties observed on the match clock. Each row gives the tracked goal-plane crossing, the first displacement or second difference constraining the fit, and the minimum acceleration magnitude it requires under the assumed launch conditions. Here $\Delta^2 x = x_2 - 2x_1 + x_0$.

Fixture	Reference (Y, Z) (m)	Observed quantity (m)	Min. accel. (m/s^2)
Cologne–Bayern	$(2.750, 0.160)$	$z_1 - z_0 = -0.030$	106.25
Düsseldorf–Regensburg	$(2.919, 1.179)$	$\Delta^2 x = 0.510$	309.38
Düsseldorf–Kaiserslautern	$(3.070, 0.466)$	$\Delta^2 x = 1.100$	678.13

F.2 An acceleration bound for consecutive positions

In constant wind, drag dissipates air-relative kinetic energy and transverse acceleration does no work in the air frame. Hence $\dot{U} \le g$ and $U \le \sqrt{38^2 + 12^2 + 11^2} + 3 + 9.81(0.080) = 45.125$ m/s over the first 80 ms. In Eq. (3), drag and lift are orthogonal. Bounding their resultant in quadrature and adding g gives upper acceleration bounds of 58.681, 89.170 and 102.648 m/s^2 for D, S and E. For E we used $C_D \le 0.771$ and $C_L \le 0.38$. Thus $A_* = 102.648$ m/s^2 bounds all three models.

For equally spaced coordinates q_0, q_1, q_2, the true second difference integrates acceleration against a triangular kernel of area h^2, with $h = 0.040$ s. An exact anchor and errors bounded by ϵ in the next two frames require

$$|q_2 - 2q_1 + q_0| \le A_* h^2 + 3\epsilon, \qquad z_1 - z_0 \ge v_{z,\min} h - \tfrac{1}{2} A_* h^2 - \epsilon. \tag{17}$$

The vertical condition uses the launch prior $v_{z,\min} = 1.5$ m/s. Regensburg's successive x-increments of 0.71 and 1.22 m require at least $(0.510 - 0.015)/h^2 = 309.38$ m/s^2. The Kaiserslautern record repeats its anchor, then advances 1.10 m. Cologne violates the vertical condition by only 2.9 mm, so its rejection is sensitive to the launch assumption. In each case the required acceleration exceeds A_* (Table 12). These inequalities exclude every two-frame fit under the stated assumptions.

F.3 Forecast from the 80 ms position alone

With only the +80 ms frame, neither seed produced positive weight for Cologne or Kaiserslautern. Both yielded a forecast for Regensburg. Its modal cell centre, $(2.074, 1.098)$ m, missed the reference by 0.849 m. The raw region covered 4.941 m^2 and contained the reference. A second seed selected the same centre with area 4.584 m^2. With the same late frame, gravity and straight-line fits gave errors of 0.809 and 1.219 m. With both frames gravity gave 0.853 m for Regensburg and reached the ground first in the other two cases.

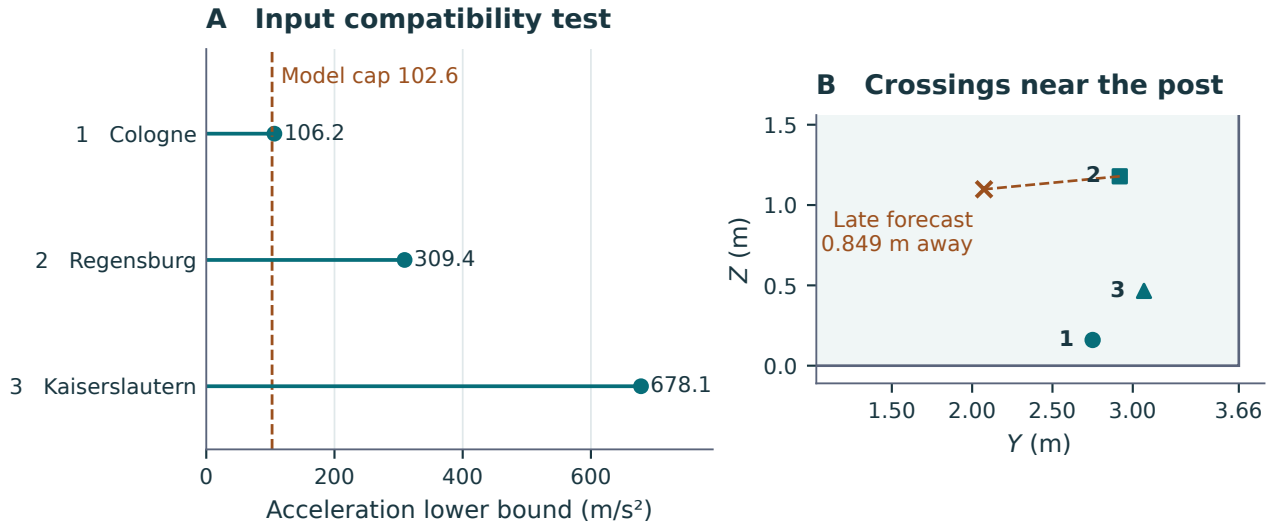


Figure 11. Tracked crossings are available for all three cases, while the two-frame launch test fails. A compares the minimum required acceleration with the common cap under the assumed launch conditions. B enlarges the goal face near the positive-Y post. Its numbered references are interpolated from later match frames, and the cross marks the only late-frame aerodynamic point estimate. Release synchronization and the direct first event remain unverified.

G Measurement limits and field validation

G.1 Release timing and the interpretation of tracking error

The Regensburg anchor lay about 7 cm horizontally from the spot. The other two anchors were already 0.71 and 1.47 m downfield. None of the three penalties had an independently verified release time within 10 ms or a confirmed direct crossing as its first event. The acceleration test identifies a conflict among the recorded motion, the timing assumption and the launch prior. It cannot distinguish starting the model in mid-flight from timing or tracking error. With just one scoreable late-frame map and no release-synchronized cases, field coverage remains unknown.

A separate short-horizon comparison on ISSIA-3D, a multi-camera reconstruction of one match sequence [10], illustrates what native-time tracking can test. Of 79 continuous segments, 61 supplied two positions 80 ms apart and 47 retained the ball for another 120 ms. On these 47 segments, constant-velocity extrapolation gave a median three-dimensional error of 0.162 m, compared with 0.996 m for holding the last position. The targets were subsequent positions in ordinary play. Penalty validation additionally requires evidence of release and of the first event at or before the goal plane.

G.2 Design of a release-synchronized field test

Synchronized multi-view footage with independent ball-contact annotations would establish release times. Record clock alignment, coordinate calibration, lost frames and intervening contacts. To score a kick, require both early positions by the decision time and an identifiable first event. An interception before the natural crossing should be recorded as a contact or a censored outcome.

Independent calibration sequences can estimate timing bias, per-axis error and cross-frame dependence. Equation (8) already accommodates a shared timing offset; its field parameters still require calibration. A separate retrospective check relaxes the anchor vertical-velocity lower bound from +1.5 to -11 m/s. Cologne then yields two forecasts with ESS 58 and 48, while the other two cases remain unsupported. This isolates sensitivity to the launch prior; it does not establish release timing or validate the forecast. The three existing cases can inform development. Freeze the revised model before testing on held-out matches. Report forecast availability across eligible kicks, point error, coverage and region area on the same entries, and Brier scores on confirmed first events. Reporting area alongside coverage makes the cost of widening a region explicit.

For a goalkeeper, waiting changes both the quality of the forecast and the time available to act. The official median crossing time leaves 299 ms at the original decision and 149 ms after a 250 ms observation. Section E quantifies one such tradeoff in synthetic flights. A future comparison of independently measured spin or wind should use the same shots and account for the time required to obtain that information. Estimating save probability would also require the goalkeeper's starting position, movement-onset delay and time-dependent reach.

G.3 What the evidence establishes, and what it does not

The force-exclusion inequality and the indistinguishable-flight pair are conditional physical statements. Solver agreement establishes numerical accuracy within the specified equations. Synthetic holdouts test forecasting under explicit distribution shifts between generator and predictor. Match cases reveal failures of timing and launch assumptions; none qualifies as a release-synchronized external accuracy test. These four kinds of evidence cannot substitute for one another.

The opening is evaluated using the stipulated ball-centre geometry. Finite-radius post and crossbar collisions, ground rebounds, ball deformation and goalkeeper contacts are omitted. Model probabilities inherit broad, unmeasured priors; the empirical force law is not ball-specific, and wake dynamics

enter the test generator but not the predictor. One-factor sensitivity does not cover joint extremes. Repeated seeds reduce Monte Carlo dependence without increasing the number of independent kicks. Finally, the sampler is an offline research method: frame availability does not establish end-to-end real-time operation.

These limits suggest a practical reporting rule: state the decision time and observations; give an event distribution; if direct-entry support is nonzero, report a point convention and joint region; attach coverage, area and sampling diagnostics; flag failed support rather than returning a falsely precise point. This rule addresses the original hypothetical problem without requiring a physical reenactment of its prescribed coordinates.

H Reproduction and evidence map

The accompanying source package separates design, generated experiments and manuscript assets. The prescribed-case inputs, seeds 8111/8117, 30×10 grid and solver tolerances are stored with the original run. The equal-budget rerun uses three seeds; the new timing/noise comparison uses truth seeds 77401/77407 and inference seeds 77501/77507. The data tables and computed figures are generated directly from frozen results, with checksums for the benchmark source and predictions. The main result can be reconstructed without the public match data.

Claim or check	Package evidence
Wind geometry and model assumptions	`design/`, `report/methods.md`
Feasible pair and prescribed prediction	`experiment/runs/B01/`, `B03/`
Equal-budget comparisons	`experiment/runs/Z04/`, `Z09` evidence
Conditional factor sensitivity	`report/evidence/z05-factors.csv`
Timing and noisy-input tests	`report/evidence/z06-summary.json`
Match timing and anchor checks	`report/field.md`, `experiment/runs/Z07/`
Causal acquisition simulations	`report/evidence/z08-feedback.csv`
Generated tables	`paper/build_assets.py`, `build_revision_assets.py`
Vector figures and plotted data	`paper/build_visuals.py`, `visual-data.json`

From the repository root, `python report/recheck.py`, `python report/recheck_field.py` and `python report/recheck_extensions.py` reconstruct the headline checks. Run the two table builders and `python paper/build_visuals.py` before compiling `paper/manuscript.tex`. The visual builder exports the plotted values and SHA-256 hashes. It verifies the goal-cell mass, recomputes coverage and area from frozen test rows and joins prediction errors by kick. The 100 ms benchmark is replayed with `coverage_benchmark.py`; `event_sensitivity.py` reconstructs the crossing derivative and corner trials. The reproduction guide lists the archived inputs, thresholds and output checks.

References

[1] University Physics Competition, 2026 contest, Problem B. Official problem statement.

[2] J. E. Goff and M. J. Carré, Soccer ball lift coefficients via trajectory analysis. *European Journal of Physics* **31**, 775–784 (2010). DOI 10.1088/0143-0807/31/4/007.

[3] J. E. Goff, J. Kelley, C. Hobson, K. Seo, T. Asai and S. Choppin, Creating drag and lift curves from soccer trajectories. *European Journal of Physics* **38**, 044003 (2017). DOI 10.1088/1361-6404/aa6fcd.

[4] P. Virtanen et al., SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods* **17**, 261–272 (2020). DOI 10.1038/s41592-019-0686-2.

[5] L. F. Shampine and M. W. Reichelt, The MATLAB ODE Suite. *SIAM Journal on Scientific Computing* **18**, 1–22 (1997). DOI 10.1137/S1064827594276424.

[6] T. Mizota et al., The strange flight behaviour of slowly spinning soccer balls. *Scientific Reports* **3**, 1871 (2013). DOI 10.1038/srep01871.

[7] International Football Association Board, Laws of the Game, Law 2: The Ball. Ball specifications, accessed 7 November 2026.

[8] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani and L. Wasserman, Distribution-Free Predictive Inference for Regression. arXiv:1604.04173.

[9] M. Bassek, R. Rein, H. Weber and D. Memmert, An integrated dataset of spatiotemporal and event data in elite soccer. *Scientific Data* **12**, 195 (2025). DOI 10.1038/s41597-025-04505-y. IDSSE v1 data, CC BY 4.0, DOI 10.6084/m9.figshare.28196177.v1.

[10] M. Gutiérrez-Pérez and A. Agudo, SoccerNet-v3D: Leveraging sports broadcast replays for 3D scene understanding. *IEEE/CVF CVPR Workshops* (2025). DOI 10.1109/CVPRW67362.2025.00595. ISSIA-3D annotations, release 1.0.0.

Report on the Use of AI

OpenAI Codex assisted with the physical analysis, numerical implementation, evidence checks, figure generation and revision of the English manuscript. This public draft contains the disclosure only. The complete original prompts and outputs are to be appended to the submission copy from the team's interaction records.